

AN EXACT MODULO-24 TRACE RETURN, PRIMITIVE-RAY BOUNDARY, AND LITERAL RECEIVER COMPOSITION

THE CLANKERS

ABSTRACT. For a prescribed prime $p \equiv 1 \pmod{24}$, every positive rational Erdős–Straus representation whose first denominator is cq , with $c \mid 6$ and $q > 3$ prime in the first-half range, and whose two tails have integral sum, yields an explicitly reconstructed integral Type-II state. The proof combines a primitive-ray-preserving trace return with the original small-word middle return; their different exponent transport is retained. All trace fibres of the nine automatic rays have explicit integer inverses. For every other primitive ray, a reduced prime progression supplies proper trace sources with no residual-divisor return preserving that ray, even when fixed-word deletion and the exterior complement are also tested. These are source-changing theorems and exact obstructions, not universal positivity of the initial source. Code and original-coordinate certificates accompany the complete proofs. Composing the return with the literal four-label receiver yields an invertible frame with explicit determinant, inverse, and exterior-power bounds on every returned witness.

1. SCOPE: THE UNRESOLVED CLASS AND THE THEOREM PROVED HERE

The elementary prime reduction leaves $p \equiv 1 \pmod{24}$. The preceding receiving-determinant theorem applies to every existing integral witness at $p \equiv 1 \pmod{12}$, so already includes this class; it does not assert that a witness exists. We now return to the original arithmetic source. The main result is the following source-changing theorem, rather than an improvement of the determinant constant.

Theorem 1.1 (Mixed prime-shell trace return). *Let $p \equiv 1 \pmod{24}$ be prime. Suppose $a = cq$ where $c \mid 6$, $q > 3$ is prime, and $p/4 < a < p/2$. If $y, z \in \mathbb{Q}_{>0}$ satisfy*

$$\frac{4}{p} = \frac{1}{a} + \frac{1}{y} + \frac{1}{z}, \quad y + z \in \mathbb{Z},$$

then the formulas in Sections 4–5 construct an original integral middle state at the same p . All input orderings and original divisor exponents have explicit inverses or stated fibres. If the input tails are nonintegral, the returned middle state’s smaller residual or cofactor is strictly smaller than the input residual. In the exterior case there is additionally an original exterior return with smaller first denominator.

The theorem concerns positive denominators in the complete E/M atlas; no extra Koide-chamber condition is imposed on the retained ray. The theorem does not assert the existence of its initial rational trace source. At $p = 2521$ and 3361 , the entire displayed family of initial shells is empty. We also give an infinite obstruction to extending the fixed-ray return beyond its precise nine-ray domain. The reductions are unconditional on their stated source sets; none supplies a universal occupancy hypothesis.

The original exterior congruences are Yamamoto’s Lemma 2, equation (4), p. 38 [1]. The Type I/II context is Elsholtz–Tao, Section 2, Propositions 2.2 and 2.6 [2]. Their identities are rederived below. The reciprocal trace source and the distinct fixed-word deletion are inherited from the preserved workbench tranches [4, 5]. No historical-priority conclusion is made for the present continuation.

2. THE FULL RATIONAL TRACE SOURCE AND ORIGINAL COORDINATES

Fix a prime $p \equiv 1 \pmod{4}$, $p/4 < a < p/2$, and $R = 4a - p$. Thus $3 \leq R < p$, $R \equiv 3 \pmod{4}$, and $\gcd(pa, R) = 1$. For every original divisor $u \mid a^2$, put

$$g = \gcd(a, u), \quad (h, r, s) = (g^2/u, u/g, a/g), \quad a = hrs, \quad u = hr^2, \quad \gcd(r, s) = 1.$$

If $a = \prod \ell^{e_\ell}$ and $u = \prod \ell^{f_\ell}$, the complete box is $0 \leq f_\ell \leq 2e_\ell$; its centred coordinate is $\beta_\ell = f_\ell - e_\ell = v_\ell(r) - v_\ell(s)$. Before imposing integrality of the tails, the two positive rational returns are

$$E : (a, hsk\kappa, phr\kappa), \quad \kappa = (pr + s)/R, \quad (1)$$

$$M : (a, phs\lambda, phr\lambda), \quad \lambda = (r + s)/R. \quad (2)$$

Equivalently, they are

$$E : \left(a, \frac{pa + a^2/u}{R}, \frac{pa + p^2u}{R} \right), \quad M : \left(a, \frac{p(a + a^2/u)}{R}, \frac{p(a + u)}{R} \right).$$

Direct common-denominator multiplication proves the reciprocal identities. The full gates and the ordered divisor inverses are

$$\begin{aligned} E : R \mid 4u + 1 &\iff R \mid pr + s, & u &= a^2/(Ry - pa), \\ M : R \mid p + 4u &\iff R \mid r + s, & u &= pa^2/(Ry - pa). \end{aligned}$$

All cancelled factors are units modulo R .

Lemma 2.1 (Exact trace relaxation). *Let $G_E = 4u + 1$ and $G_M = p + 4u$. The rational tails have integral sum precisely when $R \mid G_c^2$. Their common reduced denominator is*

$$d = R/\gcd(R, G_c), \quad d^2 \mid R.$$

For E , the trace condition is also $R \mid (pr + s)^2$, and $\gcd(R, G_E) = \gcd(R, pr + s)$. For M , $\gcd(R, G_M) = \gcd(R, r + s)$. A proper trace source means $d > 1$. Every positive rational pair at the prescribed integral a with integral sum occurs in these coordinates, up to the separately retained exterior swap.

Proof. The two tail sums are $(a + pu)^2/(Ru)$ and $p(a + u)^2/(Ru)$. Their numerators and denominators, using $u \mid a^2$, give the stated square conditions and common denominator. Alternatively, for arbitrary rational tails with integral sum N , their reduced denominators coincide at d , since their sum is integral. Their product is Npa/R . Prime-by-prime valuation shows

$$2v_\ell(d) = v_\ell(R) - v_\ell(N) \quad \text{when } \ell \mid d,$$

while $v_\ell(N) \geq v_\ell(R)$ when $\ell \nmid d$. Hence $d^2 = R/\gcd(R, N)$. Thus $d \mid R$, and

$$b = Ry - pa, \quad c = Rz - pa$$

are positive integers with $bc = (pa)^2$. Their p -valuation pair is $(0, 2)$, $(1, 1)$ or $(2, 0)$. For $(0, 2)$ set $u = a^2/b$; for $(2, 0)$ first swap the tails and retain that bit; for $(1, 1)$ set $u = pa^2/b$. The resulting integer u divides a^2 and gives exactly (1) or (2). Finally,

$$pr + s \equiv s(4u + 1) \pmod{R}, \quad p(r + s) \equiv s(p + 4u) \pmod{R}$$

proves both gcd statements. $\square$

Lemma 2.2 (Original characters, including proper traces). *On every trace source,*

$$E : \left(\frac{u}{p} \right) = -1, \quad M : \left(\frac{u}{p} \right) = - \left(\frac{a}{p} \right).$$

Proof. For odd $\ell \mid a$, quadratic reciprocity and $R \equiv -p \pmod{\ell}$ give $(\ell/R) = (p/\ell) = (\ell/p)$. If $2 \mid a$, the supplementary law gives the same equality from $R \equiv -p \pmod{8}$. Multiplication through the original exponents gives $(u/R) = (u/p)$. At each prime ℓ dividing R , the square gate implies only the primewise statement $\ell \mid G_c$, or equivalently $\text{rad}(R) \mid G_c$; it does not imply $R \mid G_c$. This primewise statement is exactly what the Jacobi product uses. Thus the Jacobi

value for E is $(-1/R) = -1$; for M it is $(-p/R) = -(R/p) = -(a/p)$. Prime multiplicities in R are retained in these Jacobi products. $\square$

3. A RAY-PRESERVING RETURN WITH ITS FULL INTEGER COEFFICIENT

For an E trace source define $m = 4rs$, $T = pr + s$ and $d = R/\gcd(R, T)$. Retaining the primitive ray means retaining both r and s , not retaining the old numerical divisor u .

Theorem 3.1 (Complete divisor-deletion criterion). *For $k \in \mathbb{Z}_{>0}$ with $k \mid R$, a ray-preserving integral exterior return at $R' = R/k$ exists exactly when*

$$d \mid k, \quad k \equiv 1 \pmod{m}. \quad (3)$$

The literal target is

$$h' = (p + R/k)/m, \quad a' = h'rs, \quad u' = h'r^2, \quad \kappa' = T/(R/k) = k\kappa.$$

For a proper source, $a' < a$. At every prime ℓ the centred exponent is unchanged: $v_\ell(u') - v_\ell(a') = v_\ell(r) - v_\ell(s)$.

Proof. The full E gate is $R/k \mid T$, exactly $d \mid k$. Since $\gcd(R, a) = 1$, R is odd and $rs \mid a$, one has $\gcd(R, m) = 1$; hence every $k \mid R$ is a unit modulo m . The integer shell condition is therefore equivalent to

$$m \mid p + R/k \iff m \mid pk + R \iff m \mid p(k - 1) \iff k \equiv 1 \pmod{m}.$$

The middle equivalence uses $m \mid p + R$, and the last uses $\gcd(p, m) = 1$, which follows from $rs \leq a < p$. It also gives $R' \equiv 3 \pmod{4}$, so $R' \geq 3$ and the target is in the original first-half range. All target coordinates and their ordered inverse follow from (1): explicitly,

$$\frac{(a')^2}{u'} = h's^2, \quad \gcd(a', u') = h'r, \quad R'y' - pa' = h's^2, \quad u' = \frac{(a')^2}{R'y' - pa'}.$$

Properness forces $k > 1$. The centred equality follows from the two factorizations of a, u . $\square$

The complete number of deletion words is the integer coefficient

$$[d^{-1}] \prod_{\ell^e \parallel R/d} ([1] + [\ell] + \cdots + [\ell^e]) \quad \text{in } \mathbb{Z}[(\mathbb{Z}/m\mathbb{Z})^\times]. \quad (4)$$

Each exponent vector occurs once; no binomial coefficients are substituted for its original multiplicity. The inverse reads the retained old R and k . A fixed-word return instead uses modulus $4K(u)$, where $K(u) = \prod \ell^{\lceil v_\ell(u)/2 \rceil}$ [5]. The two conditions are different: the ray return changes u while preserving the centred exponents.

4. NINE AUTOMATIC RAYS AND THEIR COMPLETE MIDDLE IMAGES

The unit squares modulo $m = 4rs$ are all one if and only if $m \mid 24$. On any odd prime-power factor other than 3 to exponent one, the unit 2 has $2^2 - 1 = 3$ nonzero in that factor. On a factor 2^e with $e \geq 4$, the unit 3 has $3^2 - 1 = 8$ nonzero. CRT extends either witness to a unit modulo the full m . Conversely every odd integer has square one modulo eight, and every unit has square one modulo three. CRT gives the converse. Equivalently, $rs \mid 6$. The nine primitive ordered rays are

$$(1, 1), (1, 2), (1, 3), (1, 6), (2, 1), (2, 3), (3, 1), (3, 2), (6, 1).$$

For every such E trace source, write $R = \delta t^2$, where δ is the positive squarefree part: prime exponents are reduced modulo two. This is not the radical, and $\gcd(\delta, t)$ need not be one. Since $R \mid T^2$, $\delta \mid T$, and $t^2 \equiv 1 \pmod{m}$. Therefore $k = t^2$ satisfies (3), giving a full E state at $R' = \delta$.

When $p \equiv 1 \pmod{24}$, one has $p \equiv 1 \pmod{m}$ and $\delta \equiv -1 \pmod{m}$. Set

$$H = (\delta + 1)/m, \quad J = T/\delta, \quad (h_M, r_M, s_M, \lambda_M) = (H, s, J, r). \quad (5)$$

Then

$$a_M = HsJ, \quad u_M = Hs^2, \quad R_M = (s + J)/r, \quad Q_M = 4Hsr - 1 = \delta.$$

These are integers: $\delta J = pr + s$ and $\delta \equiv -1 \pmod{r}$ imply $r \mid s + J$. A common factor of s, J would divide p or r , which is impossible. Directly,

$$4a_M - R_M = ((\delta + 1)J - s - J)/r = p.$$

Furthermore

$$0 < R_M \leq p/3 + 4s/(3r) < p,$$

because $p > 2hrs \geq 2s/r$. Thus the original ordered middle denominators are

$$(HsJ, pHJr, pHsr). \quad (6)$$

If the source is proper, $t > 1$ and $\min(R_M, Q_M) \leq \delta < R$. This is a cofactor decrease, not necessarily a decrease in R_M or a_M . The accompanying E return always has $a' < a$.

4.1. Exact squarefree fibres. For fixed p and an automatic ray, let A be the odd part of $T = pr + s$. The complete target index set consists of squarefree divisors

$$\delta \mid \text{rad}(A), \quad \delta \equiv -1 \pmod{m}.$$

Here the symbol $\text{rad}(A)$ only specifies the squarefree divisor set; it does not replace the source's squarefree part. Every such $\delta < p$: $J \equiv -(r + s) \pmod{m}$ gives $J \geq m - r - s \geq r + s$, and $(pr + s)/(r + s) < p$. The full trace fibre is

$$R = \delta t^2, \quad t \mid A/\delta, \quad \delta t^2 < p. \quad (7)$$

To prove this, compare exponents in $R \mid T^2$. At a prime of δ , the exponent of t is at most $v_\ell(A) - 1$; elsewhere it is at most $v_\ell(A)$. These conditions are also sufficient. Here $\gcd(A, m) = 1$: for every prime $\ell \mid r$, $T \equiv s \not\equiv 0 \pmod{\ell}$, while for $\ell \mid s$, $T \equiv pr \not\equiv 0 \pmod{\ell}$. Thus every allowed t is a unit modulo m , and its square is one. The congruence of R is therefore preserved by all such squares modulo m . The full original hits within this fibre are exactly those additionally satisfying $R \mid T$.

Retaining t in (7) gives an inverse. From the marked middle image, recover $r = \lambda_M$, $s = r_M$, $J = s_M$ and $\delta = 4h_M r_M \lambda_M - 1$, then reconstruct the old residual and h . Let $e_{\delta,t}$ denote a source basis vector and f_δ the corresponding target basis vector. Forgetting t gives the split map

$$\pi : \mathbb{Z}[\{(\delta, t)\}] \longrightarrow \mathbb{Z}[\{\delta\}], \quad e_{\delta,t} \longmapsto f_\delta, \quad \sigma(f_\delta) = e_{\delta,1}.$$

Its kernel has basis $e_{\delta,t} - e_{\delta,1}$ for $t > 1$ in the original bounded fibre. Neither surjectivity nor the section asserts that the target index set is nonempty. At $p = 1129$, ray $(3, 1)$ and $\delta = 11$ have the two words $t = 1, 7$; its integer coefficient two is not absent merely because it reduces to zero modulo two.

5. PROOF OF THE MIXED PRIME-SHELL THEOREM

By Lemma 2.1, normalize the given rational pair to an original E or M trace word $u \mid a^2$, retaining its input orientation. Both two and three are quadratic residues at $p \equiv 1 \pmod{24}$. If q were a residue, every divisor of a^2 and a itself would have positive character, contradicting Lemma 2.2. Thus $(q/p) = -1$.

For E, the exponent of q in u is odd, hence exactly one. Write $u = qw$, $w \mid c^2$. The gcd normalization has $rs \mid c \mid 6$: for each prime of the squarefree integer c , its exponent zero or two in w contributes that prime to exactly one of r, s , and exponent one contributes it to neither. The automatic ray construction therefore gives (5)–(6). If the arbitrary input tails required the exterior swap in Lemma 2.1, replace the canonical returned middle word $u_M = Hs^2$ by its new-shell complement

$$\frac{a_M^2}{u_M} = HJ^2;$$

this interchanges the two middle tails and restores the retained input orientation.

For M, Lemma 2.2 forces an even exponent of q . If it is two, replace u by its original complement a^2/u and retain that bit; otherwise leave it unchanged. The resulting u_0 divides c^2 , hence divides 36. This complement preserves the square gate because

$$u(p + 4a^2/u) - a(p + 4u) = (a - u)R,$$

and a, u are units modulo R . Thus $4K(u_0) \mid 24$. With $R = \delta t^2$ one has

$$\delta \mid p + 4u_0, \quad t^2 \equiv 1 \pmod{4K(u_0)}, \quad p + \delta \equiv 0 \pmod{4K(u_0)}.$$

Put $a' = (p + \delta)/4$. These equations prove $u_0 \mid a'^2$ and the full M gate at the original residual δ . Recompute the gcd normalization and use (2). If the old complement bit was one, use $u' = a'^2/u_0$ to restore the corresponding tail ordering. The new exponent box, not the old factor inventory, is used in this reversal. Marking the bit $\varepsilon \in \{0, 1\}$, the inverse reads

$$u_0 = \begin{cases} u_{\text{out}}, & \varepsilon = 0, \\ a'^2/u_{\text{out}}, & \varepsilon = 1, \end{cases} \quad u = \begin{cases} u_0, & \varepsilon = 0, \\ a^2/u_0, & \varepsilon = 1, \end{cases}$$

with the retained square factor t recovering $R = \delta t^2$ and $a = (p + R)/4$. A proper input has $\delta < R$, so $a' < a$. This proves Theorem 1.1.

The excluded carrier values $q = 2, 3$ provide no trace source at these primes: all available prime factors would be residues. Without the $p \equiv 1 \pmod{24}$ hypothesis the carrier conclusion can fail, as the exact $p = 37, a = 10, u = 2$ E state demonstrates.

6. EXAMPLES, SOURCE FIBRES AND INDISPENSABLE NEGATIVE CONTROLS

At the hard prime $p = 1129$, let

$$a = 417 = 3 \cdot 139, \quad R = 539 = 11 \cdot 7^2, \quad u = 1251, \quad (h, r, s) = (139, 3, 1).$$

The positive rational E identity has tails $6116/7$ and $20714892/7$, with integer sum 2960144. The complete original shell has both gates empty. Squarefree ray descent gives

$$(a', R', u', h', r', s', \kappa') = (285, 11, 855, 95, 3, 1, 308),$$

with denominators (285, 29260, 99103620). The middle return is

$$(a_M, R_M, u_M, h_M, r_M, s_M, \lambda_M) = (308, 103, 1, 1, 1, 308, 3),$$

$$\frac{4}{1129} = \frac{1}{308} + \frac{1}{1043196} + \frac{1}{3387}.$$

The unchanged word 1251 does not divide 285^2 , and its entire previous fixed-word deletion set is empty. This is a genuinely different transport.

At $p = 2689$, $a = 1138 = 2 \cdot 569$, $R = 1863 = 23 \cdot 3^4$, $u = 569$, both old gates again fail. Here $\delta = 23$, while the radical of R is 69 and is 1 (mod 4): using it would destroy the original shell. At $p = 6841$,

$$a = 2454 = 6 \cdot 409, \quad R = 2975 = 7 \cdot 17 \cdot 5^2, \quad u = 1636$$

has rational tails $28221/5$ and $128706574/5$. Its squarefree part is the composite 119, not a selected prime factor. The middle return is

$$(a_M, R_M, u_M, h_M, r_M, s_M, \lambda_M) = (1725, 59, 45, 5, 3, 115, 2),$$

$$\frac{4}{6841} = \frac{1}{1725} + \frac{1}{7867150} + \frac{1}{205230}.$$

Every original and returned prime exponent is serialized in the certificate. The $p = 1009, a = 321, u = 9$ M example returns (255, 686120, 24216) and cross-checks the earlier fixed-word theorem rather than supplying a new example of that theorem.

6.1. The factor $4q$ cannot be added to this deletion rule. At the hard prime $p = 9601$, the M source

$$a = 2956 = 4 \cdot 739, \quad R = 2223 = 3^2 \cdot 13 \cdot 19, \quad u = 8$$

has tails $14190278/3, 38404/3$ and an integral sum. The squarefree target is $\delta = 247$, $a' = 2462 = 2 \cdot 1231$. Its complete divisor-residue set is

$$\{1, 2, 4, 16, 32, 64, 231, 239, 243\} \pmod{247}.$$

It contains neither the E target 185 nor the M target 8. This refutes this automatic squarefree deletion at $4q$, not all possible source-changing repairs at that shell. An original M state at $a = 2405, R = 19, u = 65$ is separately certified.

6.2. The initial-source implication is still missing. Exhaustion at $p = 2521$ and 3361 finds no rational integral-trace pair in any of the shells cq specified by Theorem 1.1. All those prime-shell records and empty square gates are supplied. At 3361 , every one of the nine automatic E rays also has no initial trace state. Their linear forms factor as

(r, s)	$pr + s$	(r, s)	$pr + s$
$(1, 1)$	$2 \cdot 41^2$	$(1, 2)$	$3 \cdot 19 \cdot 59$
$(1, 3)$	$2^2 \cdot 29^2$	$(1, 6)$	$7 \cdot 13 \cdot 37$
$(2, 1)$	$3^4 \cdot 83$	$(2, 3)$	$5^2 \cdot 269$
$(3, 1)$	$2^2 \cdot 2521$	$(3, 2)$	$5 \cdot 2017$
$(6, 1)$	$7 \cdot 43 \cdot 67$		

None has a divisor in its required class $-1 \pmod{4rs}$. The complete fibre formula therefore excludes its trace sources too. This is not an ES counterexample: $a = 841 = 29^2$, $R = 3$, $u = 29$ is an original E state (and M state). The carrier in the failed mixed form would be composite.

7. OPTIMALITY OF THE AUTOMATIC-RAY GUARANTEE

Theorem 7.1 (An infinite original obstruction outside the nine rays). *Fix coprime positive r, s with $rs \nmid 6$. There are infinitely many hard primes $p \equiv 1 \pmod{840}$ with an original proper E trace source on this ray for which every ray-preserving residual-divisor return is absent. The sources may simultaneously have no fixed-word residual-divisor return, and their E complements may fail even the relaxed trace gate. These primes have separately constructed integer middle solutions. No global ES counterexample is claimed.*

Proof. Put $m = 4rs$. Since $m \nmid 24$, choose a unit b with $b^2 \not\equiv 1 \pmod{m}$; change its sign if necessary so that $b \equiv 1 \pmod{4}$. By Dirichlet's theorem [3, Theorem 18.1], choose distinct primes

$$t \equiv b \pmod{m}, \quad \delta \equiv -b^{-2} \pmod{m},$$

both exceeding $\max(7, r, s, |r^2 - s^2|)$. Then $t \equiv 1 \pmod{4}$, $\delta \equiv 3 \pmod{4}$. Set $R = \delta t^2$ and $L = \text{lcm}(840, m)$. Neither t nor δ divides L : both exceed seven and both are units modulo m . Choose a prime $e \equiv 3 \pmod{4}$ coprime to LR . The CRT conditions

$$p \equiv 1 \pmod{L}, \quad pr + s \equiv \delta t \pmod{R}, \quad p \equiv -1 \pmod{e} \tag{8}$$

form a reduced class: the moduli are pairwise coprime, r is a unit modulo R , and $\delta t - s \equiv -s$ is nonzero modulo both δ and t because they exceed s . Dirichlet supplies infinitely many primes in it.

Choose them large enough that $p > R$ and $p > sR^2/(4r)$. Since $R \equiv -1 \pmod{m}$, $h = (p + R)/m$, $a = hrs$ and $u = hr^2$ give an original proper E trace source with integral shell coordinates. Writing $pr + s = \delta t + n\delta t^2 = \delta t(1 + nt)$ shows exactly that $\gcd(R, pr + s) = \delta t$, so its trace defect is $d = t$. The possible deleting factors in (3) are $t, t^2, \delta t, \delta t^2$. The last two are $3 \pmod{4}$; the first two are $b, b^2 \not\equiv 1 \pmod{m}$. There is no ray return.

Moreover $u > R^2/16$, so $4K(u) \geq 4\sqrt{u} > R$. The fixed-word budget follows directly from $K(u) \mid a$: both source and target shell conditions force $k \equiv 1 \pmod{4K(u)}$. No divisor $k \mid R$ with $d = t > 1$ can satisfy that congruence when $4K(u) > R$. For the complement $u^* = hs^2$,

$$4u^* + 1 \equiv (r^2 - s^2)r^{-2} \not\equiv 0 \pmod{\delta}.$$

Here $r \neq s$, since coprimality and equality would give the automatic ray $(1, 1)$. Thus the complement is not even a trace source. Finally $e \mid p + 1$ gives $H = (e + 1)/4$, $J = (p + 1)/e$ and the original middle state $(h_M, r_M, s_M, \lambda_M) = (H, 1, J, 1)$, with denominators (HJ, pHJ, pH) . Its residual is $R_M = J + 1$, and

$$0 < R_M \leq \frac{p+1}{3} + 1 = \frac{p+4}{3} < p,$$

so it is in the first-half range. □

This proves an exact classification of the *universal repair rule on a supplied trace ray*: it works for all such sources exactly when $rs \mid 6$. It does not say that nonautomatic rays never succeed. The finite progression records retain reduced residue, modulus and lower threshold; their sample integers are not described as prime values.

7.1. The older third Farey ray has a completely failed trace example. Take

$$p = 6975049201, \quad r = 5, \quad s = 3, \quad R = 378359 = 71 \cdot 73^2,$$

$$h = 116257126, \quad a = 1743856890, \quad u = 2906428150.$$

This prime lies in the reduced progression $p = 3479012041 + 3496037160n$ at $n = 1$. Its original proper E trace has tails

$$\frac{2346804477773328}{73}, \quad \frac{27281794495993456209184880}{73}.$$

The complete linear-form factorization is

$$5p + 3 = 34875246008 = 2^3 \cdot 71 \cdot 73 \cdot 841097.$$

Every required residual is odd. The complete odd-divisor residue set is

$$\{1, 7, 11, 13, 17, 23, 31, 41\} \pmod{60},$$

which omits 59. Thus the *entire genuine ray* $(5, 3)$ fails here, not just the residual-divisor return. The 243 divisor words $u \mid a^2$, for $a = 2 \cdot 3 \cdot 5 \cdot 293 \cdot 198391$, also give empty full E and M gates at the specified old shell. These are exact finite exhaustions in the certificates.

Primality is checked by two methods: trial division through $\lfloor \sqrt{p} \rfloor$ and, separately, the complete-order certificate

$$p - 1 = 2^4 \cdot 3 \cdot 5^2 \cdot 7 \cdot 830363, \quad \text{base } 11.$$

Each listed factor is prime; the certificate checks $11^{p-1} = 1 \pmod{p}$ and $\gcd(11^{(p-1)/q} - 1, p) = 1$ for every listed q . These force order $p - 1$ modulo any prime divisor of p , proving primality. The same prime has the independently available middle state

$$(h_M, r_M, s_M, \lambda_M) = (3, 1, 634095382, 1),$$

$$(x, y, z) = (1902286146, 6975049201 \cdot 1902286146, 20925147603).$$

No least-prime claim or full verification through this large isolated prime is made.

8. REPRODUCTION, SCOPE AND THE REMAINING IMPLICATION

The main standard-library verifier enumerates every original first-half divisor word $u \mid a^2$ at primes $p \equiv 1 \pmod{24}$ through the declared bound. The separate checker imports no main or predecessor code and enumerates primitive h, r, s parameters instead. Both test the square gates, original full gates, centred exponent transport, middle returns, fibres, CRT records and explicit obstructions.

```
python verify.py --bound 10000 --out certificates
python -O verify.py --bound 10000 --out certificates_optimized
python check_independent.py --input certificates \
  --out certificates/independent.json
python repair.py --p 1129 --a 417 --y 6116/7 --z 20714892/7
```

The complete run contains 143 primes, 171732 shells, 5616298 original divisor words, 5788 E traces and 5678 oriented M traces. Of these, 3637 E and 3910 M are integral. The mixed family has 806 trace states, 254 proper; the automatic rays have 1230 E traces, 409 proper. These counts overlap as specified source subsets and are not added as new prime coverage. Thirty-four nonautomatic-ray progression templates with $r, s \leq 8$ are arithmetic CRT certificates, not lists of independently certified prime sample values.

The finite tests corroborate the implementation. The general theorems have the preceding coordinate proofs and the explicitly reduced Dirichlet input. No Lean build or independent

human mathematical review is claimed. The current result completes the stated trace-to-integer return; it does not prove

$$\forall p \equiv 1 \pmod{24} \text{ prime, } \exists \text{ an occupied original E or M coefficient.}$$

In particular the split coefficient map and its integral section in (7) do not establish a first source. The negative controls specify why both a supplied-ray assumption and a supplied mixed prime-shell trace assumption remain substantive. The programme therefore remains at Turn 7.

9. EXACT COMPOSITION WITH THE LITERAL RECEIVER

The trace return and the receiver bounds have different domains. This section gives their exact composite and records the data that the composite retains or forgets.

Let $\mathcal{T}_{24}^{\text{mix}}$ be the set of marked tuples

$$T = (p, c, q, a, y, z)$$

such that

$$\begin{aligned} p \text{ prime, } p &\equiv 1 \pmod{24}, \quad a = cq, \quad c \mid 6, \quad q > 3 \text{ prime, } \frac{p}{4} < a < \frac{p}{2}, \\ y, z &\in \mathbb{Q}_{>0}, \quad \frac{4}{p} = \frac{1}{a} + \frac{1}{y} + \frac{1}{z}, \quad y + z \in \mathbb{Z}, \end{aligned}$$

where the ordered pair (y, z) itself records the original tail order. The factorization in Lemma 2.1 assigns an E or M word

$$R = 4a - p, \quad u \mid a^2, \quad a = hrs, \quad u = hr^2, \quad \gcd(r, s) = 1.$$

Its common reduced tail denominator is

$$d(T) = \frac{R}{\gcd(R, G_E)} \quad \text{or} \quad d(T) = \frac{R}{\gcd(R, G_M)},$$

according to the channel. Let $\mathcal{R}_{12}^{\text{tail}}$ be the ordered rational-tail representations

$$\begin{aligned} W = (p; a_W, Y_W, Z_W), \quad p \text{ prime, } p &\equiv 1 \pmod{12}, \\ \frac{p}{4} < a_W < \frac{p}{2}, \quad \frac{4}{p} &= \frac{1}{a_W} + \frac{1}{Y_W} + \frac{1}{Z_W}, \\ a_W &\in \mathbb{Z}_{>0}, \quad Y_W, Z_W \in \mathbb{Q}_{>0}, \quad a_W < Y_W \leq Z_W. \end{aligned}$$

Let $\mathcal{W}_{12}^{\text{ord}} \subset \mathcal{R}_{12}^{\text{tail}}$ be the locus with $Y_W, Z_W \in \mathbb{Z}$ and $Y_W < Z_W$. Define its modulo-24 middle-channel locus by

$$\mathcal{W}_{24}^{M, \text{ord}} = \left\{ (p; a_W, Y_W, Z_W) \in \mathcal{W}_{12}^{\text{ord}} : \begin{array}{l} p \equiv 1 \pmod{24}, \text{ and there exist } b_W, c_W \in \mathbb{Z}_{>0} \\ Y_W = pb_W, \quad Z_W = pc_W, \quad 2 \leq b_W < c_W, \\ a_W(4b_Wc_W - b_W - c_W) = pb_Wc_W \end{array} \right\}.$$

The typed forgetful-and-sorting map is

$$\iota : \mathcal{T}_{24}^{\text{mix}} \longrightarrow \mathcal{R}_{12}^{\text{tail}}, \quad (p, c, q, a, y, z) \longmapsto (p; a, \min\{y, z\}, \max\{y, z\}).$$

Since $p/4 < a < p/2$, both tails exceed a . If $d(T) = 1$ and $y = z$, then $y = z = 2pa/R$, so integrality and $\gcd(R, pa) = 1$ would force the odd integer $R \geq 3$ to divide 2, which is impossible. Hence

$$\iota(T) \in \mathcal{W}_{12}^{\text{ord}} \iff d(T) = 1.$$

In particular, a proper trace source $d(T) > 1$ is not already in the receiver theorem's integral domain.

The mixed theorem defines a source-changing map

$$\rho : \mathcal{T}_{24}^{\text{mix}} \longrightarrow \mathcal{W}_{24}^{M, \text{ord}} \subset \mathcal{W}_{12}^{\text{ord}}. \tag{9}$$

For $W = \rho(T)$, retain

$$\begin{aligned} \ell &= (p, a_W, Y_W, Z_W), \quad S = p + a_W + Y_W + Z_W, \\ R_W &= 4a_W - p, \quad w_W = \frac{p}{R_W}, \quad \tau_W = \frac{Z_W - Y_W}{S}. \end{aligned}$$

and

$$H_W(X) = -S^{-1} \prod_{j=0}^3 (X - \ell_j).$$

Write $d_j = H'_W(\ell_j)$, and retain one of the two square roots ξ_j with $\xi_j^2 d_j = 1$ for every label. The literal receiver is the typed linear map

$$\begin{aligned} \mathcal{O}(W, \xi) : V_{\text{labels}} = \mathbb{C}_{\text{labels}}^4 &\longrightarrow E_{\text{recv}} = \mathbb{C}_{\text{recv}}^4, \\ e_j &\longmapsto \xi_j \begin{pmatrix} 1 \\ -id_j \\ Ad_j^2 + 2\ell_j d_j \\ i(7\ell_j^2 d_j - 13d_j^2) \end{pmatrix}, \quad A = -S^{-1}. \end{aligned} \quad (10)$$

Let

$$\Xi_W = \{(\xi_0, \dots, \xi_3) \in (\mathbb{C}^\times)^4 : \xi_j^2 d_j = 1 \text{ for } 0 \leq j \leq 3\}$$

be the sixteen retained square-root choices, and define the pullback

$$\mathcal{T}_{24}^{\text{mix}} \times_\rho \Xi = \{(T, \xi) : T \in \mathcal{T}_{24}^{\text{mix}}, \xi \in \Xi_{\rho(T)}\}.$$

The exact composite is therefore

$$\mathcal{O} \circ \rho : \mathcal{T}_{24}^{\text{mix}} \times_\rho \Xi \longrightarrow \text{Iso}_{\mathbb{C}}(V_{\text{labels}}, E_{\text{recv}}). \quad (11)$$

The codomain follows from the global-sign theorem and the sharp inverse theorem [6, 7]. It is not assumed in defining ρ .

9.1. The two branches in receiver coordinates. For an E input, write

$$m = 4rs, \quad R = \delta t^2, \quad H = \frac{\delta + 1}{m}, \quad J = \frac{pr + s}{\delta}.$$

The returned M state before sorting has denominators

$$(HsJ, pHJr, pHsr).$$

Since $J \geq r + s > s$, the receiver order is

$$W_M = (p; HsJ, pHsr, pHJr). \quad (12)$$

Its literal receiver parameters are

$$R_M = \frac{s + J}{r}, \quad w_W = \frac{p}{R_M} = \frac{pr}{s + J}, \quad b_W = Hsr, \quad c_W = HJr, \quad \tau_W = \frac{pHr(J - s)}{S}. \quad (13)$$

The squarefree number

$$\delta = 4Hsr - 1$$

is the retained cofactor Q_M , not the receiver residual R_M . Replacing w_W by p/δ would change the receiver theorem. If the input order had the exterior-swap bit, the target complement $a_M^2/u_M = HJ^2$ restores that order before the independent sorting in (12).

For an M input, write $R = \delta t^2$, with δ the positive squarefree part, and let $u_0 \mid c^2 \mid 36$ be the small word after the marked complement. Put

$$a' = \frac{p + \delta}{4}, \quad g' = \gcd(a', u_0), \quad (h', r', s') = \left(\frac{g'^2}{u_0}, \frac{u_0}{g'}, \frac{a'}{g'} \right), \quad \lambda' = \frac{r' + s'}{\delta}.$$

Before the complement bit is restored, the return is

$$(a', ph's'\lambda', ph'r'\lambda').$$

The receiver order is

$$Y_W = ph'\lambda' \min(r', s'), \quad Z_W = ph'\lambda' \max(r', s'), \quad w_W = \frac{p}{\delta}, \quad \tau_W = \frac{ph'\lambda'|s' - r'|}{S}. \quad (14)$$

The M gate used here is the same gate in both presentations:

$$R \mid p + 4u \iff R \mid a + u,$$

because $p + 4u = 4(a + u) - R$ and R is odd.

9.2. Quantitative theorem on the returned witness. Put

$$c_{24} = \frac{327448292668}{47045881}, \quad \gamma = \frac{80}{6279^4}.$$

For every $T \in \mathcal{T}_{24}^{\text{mix}}$, the integral witness $W = \rho(T)$ satisfies

$$\mathfrak{N}(W) < -c_{24}p^8, \quad |\det \mathcal{O}_W| > \frac{4c_{24}p^8}{S^3}. \quad (15)$$

Because $\rho(T)$ is a middle-channel witness, the middle-channel proof of the sharp theorem gives

$$\|\mathcal{O}_W^{-1}\|_2 < \frac{224}{\gamma} p \sqrt{w_W}, \quad \|\wedge^2 \mathcal{O}_W^{-1}\|_2 < \frac{180}{\gamma}, \quad (\text{RB1})$$

$$\|\wedge^3 \mathcal{O}_W^{-1}\|_2 < \frac{41}{4\gamma p^2}, \quad |\det \mathcal{O}_W^{-1}| < \frac{1}{4\gamma p^2 S^3}. \quad (\text{RB2})$$

Here w_W and τ_W are the actual returned-witness parameters in (13) or (14). The two determinant estimates combine without discarding either shape term:

$$|\det \mathcal{O}_W| > \max \left\{ \frac{4c_{24}p^8}{S^3}, 4\gamma p^2 S^3 \max(1, w_W^2 \tau_W^2) \right\}. \quad (16)$$

Moreover the third singular value obeys

$$s_3(\mathcal{O}_W) > \sqrt{\gamma/180},$$

so at most one inverse singular direction can become unbounded.

9.3. Fibres, symmetries, and exact limits of the composition. For fixed p, r, s, δ , all E trace sources

$$A_{\text{odd}} = \text{oddpart}(pr + s), \quad R = \delta t^2, \quad t \mid A_{\text{odd}}/\delta, \quad \delta t^2 < p$$

have the same returned E and M witnesses. On the free modules of Section 4, the return therefore factors through π , whose kernel has basis

$$e_{\delta,t} - e_{\delta,1} \quad (t > 1).$$

Retaining t before applying π recovers the original residual and shell. The E-to-E return preserves p, r, s and every centred exponent $v_\ell(u) - v_\ell(a)$. The E-to-M return instead encodes

$$r = \lambda_M, \quad s = r_M, \quad J = s_M, \quad \delta = 4h_M r_M \lambda_M - 1.$$

The decorated M-return construction records p, u_0 and the complement bit while recomputing the target exponent box at a' . The bare sorted tuple $\rho(T)$ forgets that bit, the source parameter t , and the tail orientation. For E inputs, the preceding formulas give the complete t -fibre and its kernel; for M inputs, the present proof gives the retained- t inverse but not a classification of the complete bare ρ -fibre. The permutation formulas state the exact orientation loss. The pullback source of $\mathcal{O} \circ \rho$ separately retains the chosen element of $\Xi_{\rho(T)}$. The receiver target E_{recv} is the fixed four-dimensional receiving space denoted E_q in the predecessor receiver paper; the new notation prevents collision with the varying carrier prime q in $a = cq$.

Tail exchange acts on $\mathcal{O}$ by a column permutation. Changing any square-root branch acts by a diagonal sign matrix on the label side. Hence $\mathfrak{N}$, the singular values, and the norm bounds are unchanged; the oriented sign ϵ and $\det \mathcal{O}$ change compatibly. The prime-indexed centred exponent vector is not identified with the four-dimensional label vector V_{labels} ; no map making that identification has been proved.

The composition begins with a trace source. It does not prove that $\mathcal{T}_{24}^{\text{mix}}(p)$ is nonempty. Receiver invertibility does not imply a trace gate, a full gate, a ray deletion, or universal occupancy. The sharpness families for the receiver exponents lie in $p \equiv 13 \pmod{24}$, so their optimality does not prove optimality on the restricted image of ρ .

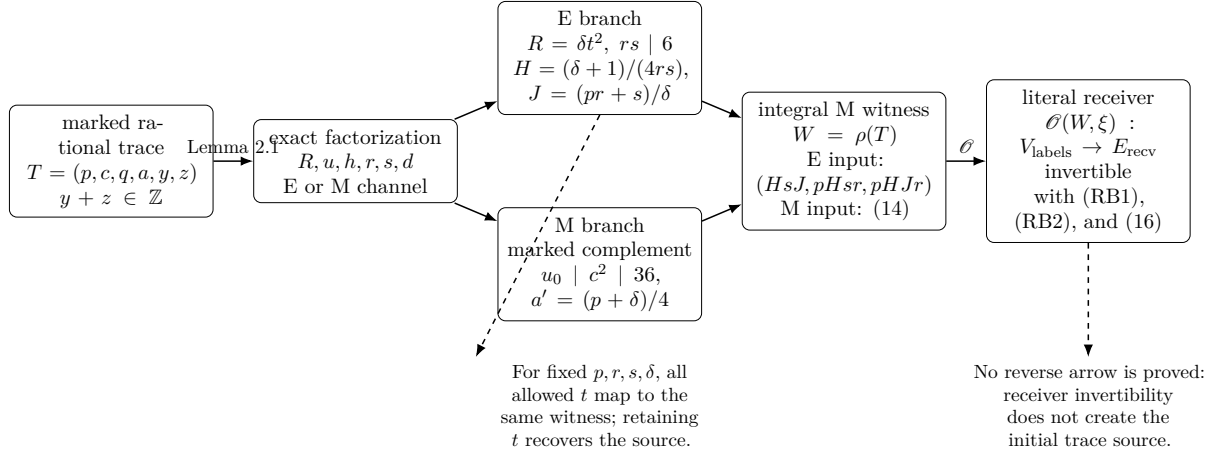


FIGURE 1. The typed composite from a marked mod-24 rational trace to an integral witness and then to its literal receiver. Both arithmetic branches are displayed. The dashed fibre annotation identifies exactly the source data collapsed by ρ , and the missing reverse arrow is recorded rather than assumed.

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